

Charizard's Chase: A WebGL Mini-Game

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Project Direction

Pokemon Mini Game

Idea Description

This project is a short Pokémon-inspired mini-game developed using WebGL with 2.5D/3D graphics. The game begins with a stylized scene where Charizard attempts to use a lighter, but a Murkrow steals it, triggering a fast-paced aerial chase. The player controls Charizard in a flying obstacle-avoidance sequence across multiple waves, avoiding objects and navigating the environment. After retrieving the lighter, the game concludes with a humorous ending where Charizard uses his own flame instead, showcasing graphical effects. The primary focus of the project is real-time rendering, lighting, and advanced visual techniques.

Baseline Techniques

- Phong Lighting Model for realistic shading
- Texture Mapping for environment and character surfaces
- Multiple Light Sources (directional light and point light for fire)
- Camera System (third-person dynamic chase camera)

Advanced Techniques

- Toon Shading (3/5): Used to achieve a stylized, cartoon-like Pokémon aesthetic
- Bloom (4/5): Used to enhance glow effects, especially for fire and highlights
- Shadow Mapping (4/5): Used to generate real-time shadows for characters and environment

Total Effort Score: 11/10

Shadow Mapping Description

Shadow mapping will be implemented by rendering the scene from the light's perspective to generate a depth map. This depth map will then be used during the main rendering pass to determine whether a fragment is in shadow. Light-space transformations will be applied to convert world coordinates into light coordinates. To handle shadow artifacts such as shadow acne, bias techniques and depth comparison adjustments will be used.

Inspiration Description

The visual style is inspired by modern Pokémon games that use toon shading for a stylized animated appearance. The gameplay draws from endless runner and flying chase mechanics, emphasizing forward motion and obstacle avoidance. The environment is influenced by low-poly stylized landscapes with bright colors and simple geometry, ensuring both visual clarity and efficient rendering.

Implementation Plan

- Week 1: Setup WebGL pipeline, shaders, and basic environment (terrain and skybox)
- Week 2: Import and optimize low-poly models, implement camera and controls
- Week 3: Develop chase gameplay (movement, obstacles, collision detection)
- Week 4: Implement toon shading and bloom effects
- Week 5: Implement shadow mapping and refine lighting system
- Final Days: Optimize performance, debug, and finalize scenes

External Assets

- Low-poly Charizard model (3.4k triangles) from Sketchfab
- Murkrow low-poly model (1–2k triangles) from Sketchfab
- Pikachu low-poly model from Thangs
- Environment textures and HDRI skyboxes from Polyhaven

All models will be converted to glTF format and optimized for real-time WebGL rendering. Polygon counts will be minimized, materials reduced, and textures compressed to ensure smooth performance. All external assets will be properly credited.